Niels Blom
Erasmus

Stevan Pavlović
5641217

Joshua Rooijakkers
Erasmus

Batuhan Yildiz
5533392

Introduction to Responsible AI Engineering

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TikTok's Algorithm with regards to Value Sensitive Design

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1 Introduction

In the field of social media entertainment, a development that has gotten the attention of almost everyone, is the fast rise of the social media application TikTok. The company was founded in 2016 by a Chinese company named Byte-Dance. Upon incorporation, the main idea of the application was the allowance for users to effectively share, edit, watch and create small 15-second personalized videos, as suggested by Qin et al. [2022]. The platform was received well and soon became one of the world's largest social media platforms, growing an audience size of 834 million users. The underlying factors of the business's success have drawn the attention of marketeers, technicians and psychologists, who were surprised by the sudden rise of TikTok.

One of the most characteristic features of the application is the hyper-personalized recommendation page. This page works with an AI-based algorithm, which collects data on user behavior, based on which it creates a feed of videos that suits the user's interest. This causes the user to engage more with the application, meaning more data becomes available to the algorithm, as noted by [Qian and Jain, 2022]. As a result, the algorithm generates an even more addictive feed and the cycle continues. This is not without consequence, as users become increasingly more vulnerable, in the end causing their well-being, mental health and cognitive liberty to be put at risk.

Since TikTok is a profit-oriented company, maximizing usage means maximizing profit [Mhalla et al., 2020]. This means that the more effective their algorithm, the more revenue they will generate. Therefore, it is beneficial for TikTok to view their algorithm as a neutral technology, which can be optimized indefinitely. However, this way of viewing design fails to take into consideration the complex sociotechnical system that the technology is part of. This brings with it the risk of ignoring stakeholders and value considerations while embedding the algorithm into society. Since it is evident that TikTok seems to have a conflict of interest in the matter, it is not clear if they can be trusted with dealing with these risks themselves. This raises the question: is it necessary that the optimization of the AI-based recommendation algorithm of TikTok is subject to regulatory oversight by an independent third-party entity?
To answer this question, we will assess TikTok's current design process and attitude by comparison to a framework for creating designs that consider human values, called Value Sensitive Design. In this way, the inadequacies of TikTok as the entity performing value-oriented investigations will come to light and the necessity for a regulatory oversight will become clear.

The remainder of this paper is structured as follows. Firstly, we will start by outlining TikTok's algorithm and its societal consequences in more detail. We will continue by analyzing these implications through the theoretical framework. We will describe how the algorithm can not be seen through the lens of utility only, and then we will apply Value Sensitive Design to arrive at a final conclusion on our research question. In this paper we will only argue that regulatory oversight is needed and would be beneficial to create a more value sensitive algorithm. We will not dive into the specific questions about the implementation of such a third entity and will leave this as a potential topic to be discussed in the future.

2 Societal implications of TikTok's algorithm

TikTok has embedded itself into a substantial part of current society. At first glance, for those who are not very familiar with the application, it might seem unclear how TikTok is much different than other earlier established social-media platforms. First of all, the popularity of an application like TikTok can be seen as a continued development over the last decades, where short videos and live-streams have become an important source of information provision. Besides, one could also observe that TikTok effectively extracts and combines features from other recent successful social-media applications [Bhandari, 2022]. It might then still be unclear how TikTok already acquired 2.6 billion downloads since February 2021 [Statista, 2023]. Not only does the application have a large amount of users, the users themselves spend an average amount of 95 minutes per day on the app [Media, 2023b]. The question that may arise is, how does such an algorithm attract people to spend such high average amounts of time, and keep opening the application, whilst it only offers small videos to its users? The answers are for a large part found in human behavior and brain functioning, and the way TikTok's algorithm anticipates on these.

It has been found that users generally have a preference for watching short videos over other types of social-media content [O'Hara, 2007] [Zhao, 2021]. TikTok responds to this preference in their own way, making them unique compared to other social-media platforms. By creating a feed for customers, which is an updated list of all new content, designed for specific user's interests, the application allows for repeated small satisfactions. Palawe [2023] suggests that satisfactions are most often provoked by a dopamine release, which is a neurotransmitter in the brain that stimulates reward. The dopamine release is associated with feelings of desire and pleasure. Besides current reward, the satisfaction caused by watching a small video influences future behavior of a customer too, as they stimulate the tendency to achieve more dopamine. Cao [2013] finds that in small video feeds, user satisfaction is an influence, and even the most significant variable found, for affecting the intention of usage continuance.

One of the key details about the application is that it invokes a recommendation algorithm that creates an interest-driven feed for its customers, as suggested by Media [2023a]. This means that the algorithm creates a feed, such that users are satisfied by small videos of their own interest. The algorithm, powered by AI, is customized in such detail that practically all users have a unique feed [Zhao, 2021]. The recommendation algorithm gets as input certain data that is tracked for all users. It collects, for each interaction with the application, the duration and number of times you watch a video, the videos you add to your favorites and the videos you share or dislike. The algorithm then creates a feed for the user that is based on this collected information about your interests. The feed often covers a large variety of user's interests, including fast news, funny videos, education, and small sitcoms.

After the user spends enough time on the application, the recommendation algorithm creates a more refined feed for each user. By carefully tracking the behavior of a customer, effective predictive models are used to predict interest characteristics, identity characteristics, and behavior characteristics to define a user's profile [Zhao, 2021]. Using this information, the algorithm learns more about the user, which makes it able to customize it's feed such that the expected usage continuance, in other words the time spent on the application by a user, can be further maximized.

As the algorithm works so effectively, users become increasingly more vulnerable to get soaked up into the application. At first, there is the continuous temptation to keep scrolling through the feed, as every video lies so much in a user's interest, yields such quick reward in the brain, and keeps stimulating further scrolling. Besides, there is often a convention among users that they are finally able to find a place where their personality will get accepted, or even get rewarded, by people that are similar to them. In this way, users can more easily focus on their own interests and communicate with like-minded people. But, they are also partially demotivated to keep up and create a real narrow circle with personal contacts themselves, to develop their social skills and to seek real-life experiences which may develop them as humans.

The effect on mental health, well-being and independence of this algorithm is in a lot of other ways harmful to users. An experimental study on the effect of small video content on the brain [Su, 2021] found that TikTok

usage is related to less self-control ability under young adults. The loss of self-control is in a lot of ways detrimental to users. Self-control is essential for shifting negative, anxiety-related feelings towards other emotions. Also, people that have less self-control are more sensitive to negative thoughts brought to mind by anxiety. Furthermore, around 5.9 % of all users, so approximately 50 million users, may have significant problematic use, which correlates with mental issues [Su, 2021]. Other negative effects that have found to be related to the usage of TikTok are loneliness, poor sleep quality, provoking thoughts of self-harm and higher levels of mental distress.

As suggested by Cao [2020], platforms like TikTok design an recommendation algorithm that eventually traps a lot of users in a cycle of dependence. But, as user welfare conflicts with the attention-based, profit-generating business model of TikTok, not much is done to prevent users from harm. As the recommendation algorithm seems to become increasingly more effective by rapid development in the field of AI, and the amount of users are rapidly increasing, there is reason for concern. A large part of society is vulnerable to the usage of an application that has found to be related to decreased cognitive liberty, mental health problems and other negative implications. Most users seem unaware, or helpless, against the algorithm that makes smart use of the stimulus system in the brain which constantly seeks for more reward.

3 Applying theoretical framework

Part of the reason why the AI-algorithm has been able to generate such far-reaching societal consequences is due to the perspective TikTok seems to hold in regards to its technology. Since its business model is centered around the retention of attention, the equation becomes simple. A more effective (and addictive) technology equals a greater profit for the company In this sense, TikTok regards its AI-algorithm as something completely neutral and value-free. They adopt this instrumentalist perspective where they see technology as nothing more than a tool to maximize the company-defined goal, namely revenue. However, this view of technology fails to consider ethical implications that stem from the technology being part of a sociotechnical system, a wider system that depends not only on the technology itself, but also on human behavior and social institutions [Van de Poel, 2020]. This framework outlines how technology and society can not be viewed as two different entities, but rather as co-dependent instances constantly influencing and working with each other [Ropohl, 1999]. In the case of TikTok's algorithm, the effects on mental health, well-being and independence of users, outlined in the previous section, show that the position of the algorithm within the sociotechnical system is clearly far from neutral. TikTok's failure, or unwillingness, to recognize the societal implications their technology might have, while constantly optimizing the algorithm as though it were something neutral, only further increases the consequences to the fundamental rights and values of its users. For the technology to be implemented in a responsible manner, these consequences simply can not be ignored. Therefore, a technology such as TikTok's algorithm should be designed while taking human values into account.

3.1 Why Value Sensitive Design?

Since it is clear that a technology such as TikTok's AI-based algorithm performs within a sociotechnical system and should thus be designed with consideration of human values. To gain a better understanding of how TikTok performs in this field, we can analyze its current design and attitude in relation to a framework called Value Sensitive Design. This framework, in short, outlines the process necessary to design technology that 'accounts for human values' [Friedman et al., 2002]. Comparing the two can help make the inadequacies in TikTok's current approach become more explicit and thus help display the necessity of change in its current approach.

3.2 Value Sensitive Design

Value Sensitive Design is defined as a theoretically grounded approach to the design of technology that accounts for human values in a principled and comprehensive manner throughout the design process. To do this, it makes use of a three-way methodology, consisting of conceptual, empirical and technical investigations, which are performed iteratively [Friedman et al., 2002]. For each of these types of investigations, we will outline what they constitute and how TikTok's approach compares.

3.2.1 Conceptual investigation

To be able to account for value dimensions in the design process, it is necessary to first have a clear picture of what the concept of values actually represents in the context-specific challenge at hand. This is what is done during the conceptual investigations in the framework of Value Sensitive Design. These investigations are philosophically informed analyses of the project under inspection, that intend to conceptualize values with the aim of clarifying fundamental (ethical) issues raised by the design that is being developed [Friedman et al., 2002]. This part of the investigation is not reserved for empirical research, but rather for the designers themselves to carefully consider how individuals that directly or indirectly interact with the system (stakeholders) could be socially impacted by the technology at hand. This step of Value Sensitive Design should help designers answer questions on how they view and prioritize the values to be implemented, who should be kept in mind during the development of technologies and what attitude the designers should hold in regards to the trade-off between moral and non-moral values.

TikTok's inadequacy in regards to this part of Value Sensitive Design mainly comes from a lack of consideration of that last question. The instrumentalist view of technology they adopt, focused on maximizing effectiveness of their algorithm for the sake of profit, clearly shows that there is little consideration of how the non-moral value of generating profit should stand in relation to other moral values and rights. In this case those moral values and rights relating to well-being, health and independence. The design process being so heavily focused on utility in general seems to view the conceptual investigation as something redundant. It is deemed unnecessary to conceptualize context-specific values and stakeholders in the first place, as technology in TikTok's design process only seems to exist in the model of optimization.

3.2.2 Empirical Investigation

Since conceptual investigation is philosophically grounded, it can only speculate on what the societal implications of a specific technology being implemented will be. To design responsibly, the analyses should go a step further. This is where empirical investigation comes into play. In the framework of Value Sensitive Design, empirical investigation can be used ahead of implementation to clarify the relevant social context, but also to concretely evaluate the consequences of an already implemented technology. Here, we adopt the definition that empirical investigation is 'any human activity that can be observed, measured or documented' [Friedman et al., 2002]. TikTok makes significant use of empirical investigation in the design process of their AI-based recommendation algorithm. Namely, empirical approaches such as the studying of user behavior and understanding human physiology are fundamentally integrated into the technology. However, instead of using this information to observe if the algorithm will adhere/adheres to relevant human values, as intended for Value Sensitive Design, TikTok uses this information to further optimize their algorithms.

As mentioned previously, they for instance use the extensive information available describing the human neurological system to gain an understanding of what an algorithm should do to create a tendency for dopamine, all to captivate the user's attention fully and to completely engage their time [Cao, 2020]. As another example, empirical evidence can be used to show that the young people are especially vulnerable to social media addiction [Su, 2021]. As it happens, this is exactly the group among which TikTok is most popular Hou [2018]. This raises questions about whether TikTok is empirically informed on the demographics on which its technology might operate and the vulnerability of said demographics, while still actively directing its algorithm designs to cognitively engage with vulnerable individuals, recognizing it as the most efficient approach.

In this sense, it is like TikTok is using empirical investigation for a kind of anti-Value Sensitive Design. This is to imply they misuse the investigation to further endanger fundamental values of well-being and cognitive liberty, instead of accounting for them through design.

3.2.3 Technical investigation

The last type of investigation focuses more so on the nature of the designed technology. The framework of Value Sensitive Design acknowledges that the configuration and properties of a technology in itself have major implications for which values it will be able to support and which values it will struggle to support or even undermine. Technological investigation aims to concretize firstly how a design's functioning relates to human values and secondly how it could be changed to improve in relation to those values [Friedman et al., 2002].

This compatibility between the properties of a technology and human values can be recognized within TikTok's algorithm as well. While TikTok does concretize how their design workings relate to value considerations, again they only seem to do so for their own personal non-moral values of effectiveness and profit. Subsequently, the improvement of design, making up the second part of technical investigation, only seems to be in support of these narrowly defined values, even if it renders other values (such as well-being and cognitive liberty) more difficult to achieve with the newly realized design properties. Take for instance the earlier described self-refining property of the algorithm, which essentially forms a positive feedback loop. This loop begins by giving better quality recommendations, resulting in more user engagement. As the users spend more time on the platform, more usage data becomes available, resulting in further enhancement of the recommendation quality. In terms of the value of effectiveness, this property of the algorithm is great. In contrast, this property not only fails to promote values of well-being and cognitive freedom, but actually jeopardizes them. It shows that the ineptitude of TikTok in defining values through conceptual investigation also inhibits the effectiveness of the technical investigation of the design, even leading it in a direction opposite to the principles of Value Sensitive Design.

3.3 Need for external oversight

After comparing current TikTok design and attitude in relation to the tripartite framework of Value Sensitive Design, a few things become clear. Most importantly, TikTok's AI-based algorithm fails to put the three investigation methods to correct use in their process. It can even be argued that in some ways TikTok is using investigation to stray further from a Value Sensitive Design, leaving important human values unaccounted for or even putting them in jeopardy. These inadequacies in the design process can primarily be assigned to the significant lack of conceptual investigation.

The failure to conceptualize value dimensions and trade-offs between values as something relevant for the design brings about a monopoly for utility as the only design consideration. From here, all other investigations (empirical, technical) are performed with strictly this utility in mind, which proves to be counterproductive in regards to designing for values.

As TikTok itself is a profit-centered entity, there is no incentive for them to move away from usability and effectiveness as their core objectives. This means that they themselves will never opt for a broader conceptual investigation which considers human values and stakeholders. This is the case since as soon as values such as well-being and cognitive liberty become central in design, effectiveness of the algorithm and thus user engagement will drop.

As a result, the investigations necessary for Value Sensitive Design will not be conducted properly, unless some of the control over the design is outsourced to an independent third entity. Since this entity would not have the conflict of interest TikTok has while designing, it could clearly take into consideration the recommendation algorithm as an actor in a complex sociotechnical system. From here, it can clearly define values and stakeholders, and properly regulate the other investigations necessary for Value Sensitive Design. It is difficult to imagine TikTok themselves regulating their AI-based recommendation algorithm towards a more value-considering design. As a result, user well-being, mental health and cognitive liberty will continue being endangered, until the responsibility for responsible design is outsourced.

4 Conclusion

In this paper we focused on TikTok's AI-based recommendation algorithm. We discussed its properties and the societal consequences that can be observed as a result of its implementation. As we examine these outcomes, it raises the question of whether the algorithm's operation and design should be solely entrusted to Tiktok, or if a third party should be brought in to oversee and regulate the technology in order to safeguard fundamental values like the well-being and cognitive freedom of its users. In our opinion, this third party is absolutely necessary. To argue why we take such a stance, we firstly outline that TikTok's instrumentalist perspective on its algorithm fails to consider the complex sociotechnical system in which the technology is embedded.

We then argue that because the technology is embedded into such a system, it has to take human values, such as cognitive liberty, mental health and well-being, into consideration. To clearly show TikTok's lack of reflection in creating such a design, we outline the framework of Value Sensitive Design. This framework has a tripartite investigation methodology to ensure human values are taken into account. When looking at how TikTok's design process performs with regard to these three investigation processes, it becomes clear the design fails and at times even counteracts what is necessary for Value Sensitive Design. The emphasis TikTok puts on usability makes it highly unlikely that these inadequacies will be resolved, as conceptualization of other values poses a risk for the profit-based model of the company. This means that the only way to enforce that the user's fundamental values are protected is by installing a third party that does not have this conflict of interest as a regulatory force. It is beyond the scope of this paper to discuss the exact implementation of such a third party, but our conclusion does provide lots of potential interesting topics to think about in the future. How do we ensure the third party does not have a conflict of interest? What should be the extent of the third party's regulatory influence on the company's design process? This and many more questions will need to be answered for TikTok's AI-based recommendation algorithm to become value sensitive.

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